

IP address

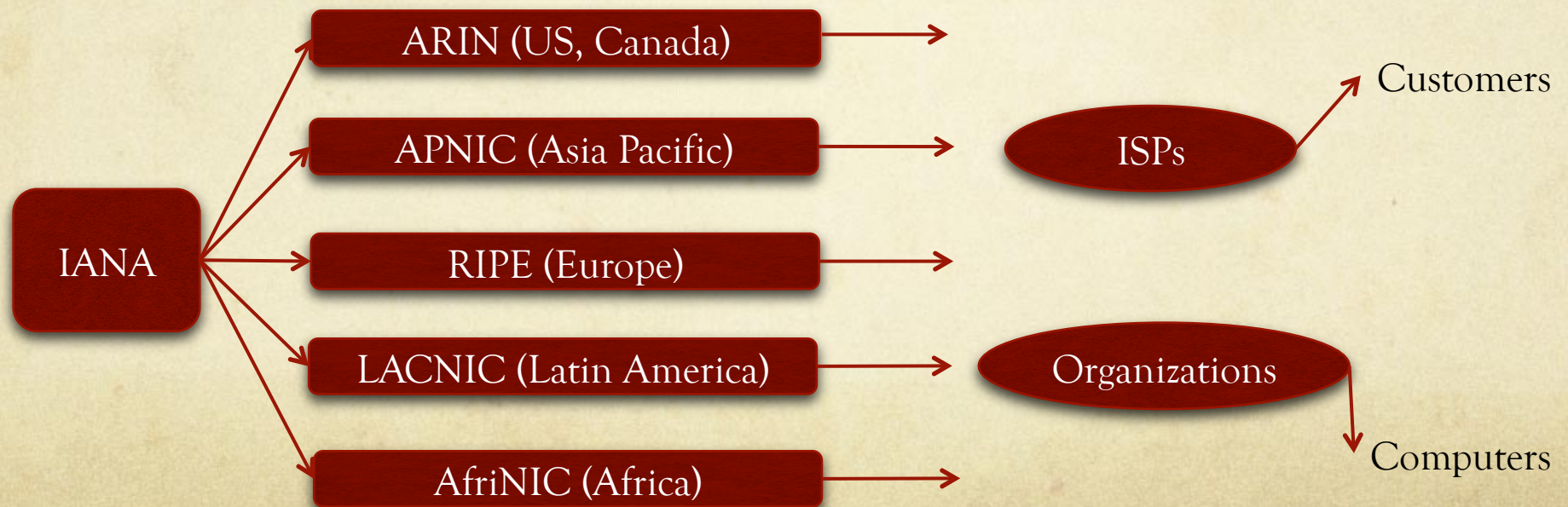
- ◆ Defining feature of IP is an IP address
 - ◆ *IPv4* → 32 bits
- ◆ IP address uniquely identifies a *network interface*
 - ◆ Host typically has *one* network interface, so *one* IP address
 - ◆ Routers have *multiple* interfaces, so *multiple* IP addresses
- ◆ IPv4 *dotted decimal* notation
 - ◆ Four 8-bit numbers (*octets*) separated by dots
 - ◆ 10101100 00010000 00000000 01100100 → 172.16.0.100

IP address

- ◆ Can be *public* or *private*
- ◆ *Public*
 - ◆ Valid destination on Internet
 - ◆ Globally known
- ◆ *Private*
 - ◆ Can be used freely *within* just private networks
 - ◆ E.g., home, small company, etc.
 - ◆ Still need *public* IP address to connect to Internet
 - ◆ Will need *NAT* to translate private to public IP addresses

IP address

- ◆ Public IP addresses are *allocated* & *managed* hierarchically
 - ◆ *IANA* delegates blocks of addresses to *regional authorities* (*RIR*)
 - ◆ *ISPs/organizations* get blocks of addresses from RIRs
 - ◆ ISPs/organizations assign addresses to *customers/computers*



IPv6 addresses

- ◆ Future of the *Internet Protocol*
 - ◆ 128-bit addresses
- ◆ IPv6 notation
 - ◆ 8 groups of 4 hexadecimal digits each separated by colons
 - ◆ 2001:0DB8:0000:0000:0000:ff00:0042:8329
 - ◆ Can omit leading zeros in any group
 - ◆ 2001:DB8:0:0:0:ff00:42:8329
 - ◆ Can omit groups of zeros altogether
 - ◆ 2001:DB8::ff00:42:8329

Issue is...

- ◆ Wide-scale *deployment* of IPv6!
- ◆ IPv6 will have to *co-exist* with IPv4 for several more years
 - ◆ *Communication* b/w computers with IPv4 & IPv6 addresses
- ◆ Some approaches have been proposed ...